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Wheel and mobility including the same

Abstract

A mobile apparatus is provided. The mobile apparatus may include: a body including a material of which a shape is reversibly deformable; a first link rotatably coupled to the body; a second link having one side rotatably coupled to the first link and the other side rotatably coupled to the body; a first shaft provided in a region, in which the first link is rotatably coupled to the body, and inserted into the first link and the body; a second shaft provided in a region, in which the second link is rotatably coupled to the first link, and inserted into the first link and the second link; and a third shaft provided in a region, in which the second link is rotatably coupled to the body, and inserted into the second link and the body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) The present application claims priority from and the benefit of Korean Patent Application No. 10-2022-0046378, filed on Apr. 14, 2022, which is hereby incorporated by reference for all

purposes.

TECHNICAL FIELD

(2) The present disclosure relates to a wheel and a mobility including the wheel.

BACKGROUND

(3) As the demands for mobile apparatuses including new types of transportation means such as delivery robots increase, research on wheels mounted on the mobile apparatuses is also being actively conducted. In particular, there are increasing demands for wheels capable of traveling smoothly while conforming to the ground or obstacles that face the mobile apparatus during the traveling of the mobile apparatus.

(4) For example, in order for the wheels to be mounted on such a mobile apparatus, each of the wheels has to be able to travel while stably supporting the mobile apparatus on a flat ground. In addition, when the mobile apparatus encounters stairs or obstacles while traveling, the shape of the wheel should be deformable in response to the shape of the stairs or obstacles. Thus, the wheel can easily climb up and down the stairs and pass over the obstacles smoothly. For this, it is preferable to apply, to the wheel, an elastic material of which the shape can be reversibly deformed by an external force.

(5) Some wheels may have an elastic material. Accordingly, such a wheel may be twisted in the width direction during the turning motion of the wheel. This acts as a factor that adversely affects the durability of the wheel, and consequently deteriorates the traveling performance of the mobile apparatus.

SUMMARY

(6) The following summary presents a simplified summary of certain features. The summary is not an extensive overview and is not intended to identify key or critical elements.

(7) Aspects of the present disclosure may relate to wheels. The wheel may comprise: a body comprising a material of which a shape is reversibly deformable; a first link rotatably coupled to the body; a second link having one side rotatably coupled to the first link and the other side rotatably coupled to the body; a first shaft disposed in a region, in which the first link is rotatably coupled to the body, and inserted into the first link and the body; a second shaft disposed in a region, in which the second link is rotatably coupled to the first link, and inserted into the first link and the second link; and a third shaft disposed in a region, in which the second link is rotatably coupled to the body, and inserted into the second link and the body.

(8) Further aspects of the present disclosure may relate to mobile apparatuses. The mobile apparatuses may comprise a wheel and a frame to which the wheel may be coupled. The wheel may comprise: a body comprising a material of which a shape is reversibly deformable; a first link rotatably coupled to the body; a second link having one side rotatably coupled to the first link and the other side rotatably coupled to the body; a first shaft disposed in a region, in which the first link is rotatably coupled to the body, and inserted into the first link and the body; a second shaft disposed in a region, in which the second link is rotatably coupled to the first link, and inserted into the first link and the second link; and a third shaft disposed in a region, in which the second link is rotatably coupled to the body, and inserted into the second link and the body.

(9) These and other features and advantages are described in greater detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the present disclosure, illustrate aspects of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

(2) FIG. **1** is a perspective view illustrating a wheel according to the present disclosure.

(3) FIG. **2** is a cross-sectional view illustrating a wheel according to the present disclosure.
(4) FIG. **3** is a perspective view illustrating a coupling structure in a wheel according to the present disclosure.
(5) FIG. **4** is a cross-sectional view illustrating another example of a coupling structure in a wheel according to the present disclosure.
(6) FIG. **5** is a plane view illustrating a body unit provided in a wheel according to the present disclosure.
(7) FIG. **6** is a view illustrating a state in which a wheel according to the present disclosure may travel along the ground.
(8) FIG. **7** is a view illustrating a state in which a wheel according to the present disclosure may travel on stairs.

DETAILED DESCRIPTION

(9) A wheel and a mobile apparatus according to the present disclosure will be described with reference to the drawings.

(10) FIG. **1** is a perspective view illustrating a wheel according to the present disclosure. Referring to FIG. **1**, a wheel **10** according to the present disclosure may include a body unit **100**. The body unit **100** may include a material, of which, a shape may be reversibly deformable. The body unit **100** may be referred to as a body or a body of a wheel. The body unit **100** may be a single component made of one material. For example, the body unit **100** may be made of a rubber material. The shape of the body unit **100** may be reversibly deformed as described herein. Accordingly, the shape may be deformed if an external force is applied thereto, but the shape may be restored to the original state if the external force is removed. Thus, the shape of the body unit **100** may be deformed to conform to the shapes of the ground or stairs when the body unit **100** receives an external force from the ground or stairs during traveling of the wheel **10**. The wheel **10** and a mobile apparatus (e.g., vehicles, robots, etc.), which may include, for example, an apparatus that is designed and constructed so as to be moveable from place to place but may not include a portable apparatus (e.g., mobile phone), provided with the wheel **10** may be stably supported while the wheel is deformed.

(11) The wheel **10** may include: a first link member **210** rotatably coupled to the body unit **100**; and a second link member **220** having one side rotatably coupled to the first link member **210** and the other side rotatably coupled to the body unit **100**. A link member (e.g., the first link member **210**, the second link member **220**, etc.) may be referred to as a link. Multiple first link members **210** and second link members **220** may be provided on a single body unit **100**. According to aspects having multiple first link members **210** and second link members **220**, the first link members **210** and the second link members **220** may be equidistantly spaced apart from each other in a circumferential direction C of the body unit **100**.

(12) FIG. **2** is a cross-sectional view illustrating a wheel according to the present disclosure. Referring to FIG. **2**, the first link member **210** and the second link member **220** may be provided on one side of the body unit **100** and face the body unit **100** in a width direction W. Therefore, the first link member **210** and the second link member **220** may support the body unit **100** in the width direction W.

(13) If the wheel **10** turns left or right while traveling, the body unit **100** may be twisted in the width direction W of the body unit **100**. In this case, the traveling performance may be significantly deteriorated during the turning motion of the wheel **10**, for example, based on the torsion of the body unit **100**. Also, the body unit **100** may be damaged when the twisting of the body unit **100** is severe. However, according to the present disclosure, the first link member **210** and the second link member **220** may support the body unit **100** in the width direction W, and thus, it is possible to minimize the twisting of the body unit **100** even during the turning motion.

(14) Still referring FIG. **2**, the wheel **10** according to the present disclosure may further include a plurality of shafts (e.g., a plurality of rotary shafts). The plurality of shafts may include a first

rotary shaft **310**. The first rotary shaft **310** may be provided in a region in which the first link member **210** is rotatably coupled to the body unit **100**. The first rotary shaft **310** may be inserted into each of the first link members **210** and the body unit **100**. The wheel **10** may further include a second rotary shaft **320** provided in a region, in which the second link member **220** is rotatably coupled to the first link member **210**. A second rotary shaft **320** may be inserted into each of the first link members **210** and the second link members **220**. The wheel **10** may further include a third rotary shaft **330** provided in a region in which the second link member **220** is rotatably coupled to the body unit **100**. The third rotary shaft **330** may be inserted into each of the second link members **220** and the body unit **100**.
(15) According to aspects of the present disclosure, the first link member **210** may be provided inward from the second link member **220** in a radial direction of the body unit **100**. Therefore, if an external force is applied to the inside of the body unit **100** in the radial direction R while the wheel **10** travels, the second link member **220**, which is positioned relatively further outside in the radial direction R than the first link member **210**, may receive the external force from the body unit **100** and may rotate relative to the first link member **210**. The rotation of the second link member **220** will be described later.
(16) FIG. **6** is a view illustrating a state in which a wheel according to the present disclosure may travel along the ground, and FIG. **7** is a view illustrating a state in which a wheel according to the present disclosure may travel on stairs.
(17) As illustrated in FIGS. **6** and **7**, when the body unit **100** is deformed while the wheel **10** travels, the second link member **220**, which may be positioned in a deformed region of the body unit **100**, may rotate while conforming to the deformed shape of the body unit **100**. FIGS. **6** and **7** illustrate an example state in which, when the body unit **100** is deformed while the wheel **10** travels, the second link member **220**, which may be positioned at a place corresponding to the deformed region of the body unit **100**, may rotate inward in the radial direction R of the body unit **100**. Accordingly, the first link member **210** and the second link member **220** may stably support the body unit **100** in the width direction W, but may not affect the shape deformation of the body unit **100** in the radial direction R due to an external force applied to the wheel **10** while it travels.
(18) As described above, the first link member **210** may be provided inward from the second link member **220** in the radial direction of the body unit **100**. Therefore, the first rotary shaft **310** may be provided inward from the second rotary shaft **320** in the radial direction R of the body unit **100**, and the second rotary shaft **320** may be provided inward from the third rotary shaft **330** in the radial direction R.
(19) Also, according to aspects of the present disclosure, the first rotary shaft **310**, the second rotary shaft **320**, and the third rotary shaft **330**, which are described above, may be provided in parallel to each other, and the first rotary shaft **310**, the second rotary shaft **320**, and the third rotary shaft **330** may be provided perpendicular to the radial direction of the body unit **100**. Therefore, a plane including a trajectory along which the first link member **210** rotates, and a plane including a trajectory along which the second link member **220** rotates, may be parallel to the radial direction R and the circumferential direction C of the body unit **100**.
(20) Referring again to FIG. **2**, according to aspects herein, the first link member **210** and the second link member **220** may be spaced apart from the body unit **100** in the width direction W of the body unit **100**. This may assist with the following advantages. In the case where the body unit **100** is twisted beyond a certain range (e.g., when the wheel **10** turns), the first link member **210** and the second link member **220** may support the body unit **100**. However, before the body unit **100** is twisted beyond the certain range (e.g., when the wheel **10** travels in a straight direction), the first link member **210** and the second link member **220** may not interfere with the body unit **100**.
(21) According to examples of the present disclosure and as illustrated in FIG. **2**, the distance between the first link member **210** and the body unit **100** may be greater than the distance between the second link member **220** and the body unit **100**. This may be understood as indicating that the

first link member **210** is biased further outward in the width direction W of the body unit **100** than the second link member **220**.

(22) During the traveling of the wheel **10**, an external force may be applied to the outermost surface in the radial direction R of the body unit **100**. Thus, a greater load due to such an external force may be applied to the second link member **220**, which may be provided relatively outside in the radial direction R, than the first link member **210**, which may be provided relatively inside in the radial direction R. Therefore, according to aspects herein, the second link member **220** may be located relatively closer to the body unit **100** in the width direction W than the first link member **210**. Accordingly, a moment arm between a point, at which the external force is directly applied to the wheel **10**, and the second link member **220** may be minimized.

(23) Alternatively, the distance between a first link member **211** and body unit **100** may be substantially the same. FIG. **4** is a cross-sectional view illustrating another example of a coupling structure in a wheel according to the present disclosure. Referring to FIG. **4** the distance between a first link member **211** and body unit **100** may be substantially the same as or correspond to the distance between a second link member **221** and a body unit **100**. According to such aspects, in a region in which a second rotary shaft (e.g., second rotary shaft **621**) passes through the first link member **211** and the second link member **221**, the first link member **211** may be inserted into the second link member **221**. In this case, a coupling structure between the first link member **211** and the second link member **221** may be symmetrical with respect to the width direction W of the body unit **100**.

(24) More specifically, as illustrated in FIG. **4**, a protrusion region **210***a*, which may protrude from the first link member **211** toward the second link member **221**, and through which the second rotary shaft **621** may pass, may be formed in the first link member **211**. Additionally, a recess region **220***a*, which may be recessed in the direction away from the first link member **211**, and through which the second rotary shaft **621** may pass, may be formed in the second link member **221**. Here, the protrusion region **210***a* may be inserted into the recess region **220***a*. However, unlike the configuration illustrated in FIG. **4**, in a region, in which a second rotary shaft **621** may pass through a first link member **211** and a second link member **221**, the second link member **221** may additionally or alternatively be inserted into the first link member **211**.

(25) FIG. **5** is a plane view illustrating a body unit **100** provided in a wheel **100** according to the present disclosure. Referring to FIG. **5**, a first body through-hole H**1**, into which the first rotary shaft **310** may be inserted, and to which the first rotary shaft **310** may be coupled may be formed in the body unit **100**. Additionally, a second body through-hole H**2**, into which the third rotary shaft **330** may be inserted, and to which the third rotary shaft **330** may be coupled, may be formed in the body unit **100**.

(26) Referring to FIG. **3**, the wheel **10** according to the present disclosure may further include a first spacer **410** which may be inserted into, and coupled to, the first body through-hole H**1** to accommodate the first rotary shaft **310**. Additionally, and a second spacer **420** which may be inserted into, and coupled to, the second body through-hole H**2** to accommodate the third rotary shaft **330**. Further, each of the first spacer **410** and the second spacer **420** may be coupled and fixed to the body unit **100**.

(27) The first spacer **410** may be configured to prevent the first rotary shaft **310** inside the first body through-hole H**1** from directly contacting the body unit **100**, and the second spacer **420** may be configured to prevent the third rotary shaft **330** inside the second body through-hole H**2** from directly contacting the body unit **100**.

(28) Referring now to FIG. **2**, the wheel **10** according to the present disclosure may further include a first bearing **510** which may be provided in a region, in which the first rotary shaft **310** may pass through the first link member **210**. First bearing **510** may be accommodated inside the first link member **210**, and the first rotary shaft **310** may pass through the first bearing **510**. Additionally, the wheel **10** according to the present disclosure may further include a second bearing **520** which may

be provided in a region, in which the second rotary shaft **320** may pass through the first link member **210**. The second bearing **520** may be accommodated inside the first link member **210**, and the second rotary shaft **320** may pass through the second bearing **520**. The first bearing **510** may be configured to support the first rotary shaft **310** so that the relative rotation between the first link member **210** and the first rotary shaft **310** occurs smoothly, and the second bearing **520** may be configured to support the second rotary shaft **320** so that the relative rotation between the first link member **210** and the second rotary shaft **320** occurs smoothly.
(29) Further, as illustrated in FIG. **2**, the first bearing **510** may include: a first bearing inner race **512** which may be coupled and fixed to the first rotary shaft **310**; and a first bearing outer race **514** which may be provided on the outside of the first bearing inner race **512** and may be coupled and fixed to the first link member **210**. Also, the second bearing **520** may include: a second bearing inner race **522** which may be coupled and fixed to the second rotary shaft **320**; and a second bearing outer race **524** which may be provided on the outside of the second bearing inner race **522** and coupled and fixed to the first link member **210**. Therefore, the first rotary shaft **310** may be supported by the first bearing **510** so as to be rotatable relative to the first link member **210**, and the second rotary shaft **320** may be supported by the second bearing **520** so as to be rotatable relative to the first link member **210**.
(30) According to aspects herein, the wheel **10** according to the present disclosure may further include one or more nut members. More specifically, as illustrated in FIGS. **2** and **3**, the wheel **10** according to the present disclosure may further include: first nut members **610** coupled to the first rotary shaft **310**; a second nut member **620** coupled to the second rotary shaft **320**, and a third nut member **630** coupled to the third rotary shaft **330**. The first to third nut members **610**, **620**, and **630** may be coupled to the first to third rotary shafts **310**, **320**, and **330**, respectively, in a bolt-nut coupling manner.
(31) The first nut members **610** (e.g., an upper first nut member and a lower first nut member as shown in FIGS. **2-3**) may be provided on each of both sides of the body unit **100** in the width direction W of the body unit **100**. The upper first nut member may be attached to the first bearing inner race **512** and/or the first bearing outer race **514**. The second nut member **620** may be provided outward from the first link member **210** in the width direction W. The third nut member **630** may face the second link member **220** with the body unit **100** therebetween in the width direction W.
(32) The first nut members **610**, second nut member **620**, and third nut member **630** may be configured to stably couple the first rotary shaft **310**, the second rotary shaft **320** and the third rotary shaft **330**, respectively, to the first link member **210** and/or the second link member **220**.
(33) Referring now to FIG. **5**, the body unit **100** may be divided into a plurality of regions. However, the plurality of regions, does not necessarily indicate that the plurality of regions are made of different materials. Rather, all regions of the body unit may be made of the same material.
(34) More specifically, the body unit **100** may include an inner region **110** that forms the inside of the body unit **100** in the radial direction R. The body unit **100** may further include an outer region **120** which may be spaced outward from the inner region **110** in the radial direction R and may form the outside of the body unit **100**. The body unit **100** may further include a plurality of spokes in a spoke region **130** which may connect the inner region **110** and the outer region **120**. Each of the inner region **110** and the outer region **120** may have a ring shape.
(35) According to the present disclosure, the first body through-hole H**1** (described above) may be formed in the inner region **110**, and the second body through-hole H**2** (described above) may be formed in the outer region **120**. Therefore, the first rotary shaft **310** may be inserted into the inner region **110**, and the third rotary shaft **330** may be inserted into the outer region **120**.
(36) According to aspects, the spoke region **130** may include a plurality of spokes along the circumferential direction C of the body unit **100**. According to aspects, the plurality of spokes of the spoke region **130** may be equidistantly spaced apart from each other in the circumferential direction C. FIG. **5** or the like illustrates, as one example, that each spoke of the spoke region **130**

may have a shape which is biased in the circumferential direction C toward the outside of the radial direction R of the body unit **100**. This may be understood as indicating that the plurality of spokes of the spoke region **130** may have a shape that is curved in one direction.

(37) According to aspects, the body unit **100** may further include a reinforcement region **140** which may be provided between the inner region **110** and the outer region **120** in the radial direction R and connect the plurality of spokes of the spoke region regions **130**. Similar to the inner region **110** and the outer region **120**, the reinforcement region **140** may have a ring shape.

(38) Referring to FIG. **1**, when an external force is not applied to the wheel **10**, that is, in a state in which the shape of the body unit **100** is not deformed, a certain angle may be formed between a longitudinal direction of the first link member **210** and a longitudinal direction of the second link member **220** in the region in which the first link member **210** is coupled to the second link member **220**. Thus, in the present concept each of the first link member **210** and the second link member **220** may not be located in parallel with the radial direction R. This may assist in accomplishing the following. When an external force is applied to the body unit **100** in the radial direction R while the wheel **10** travels, the directions in which the forces are applied to the first link member **210** and the second link member **220**, and the directions in which the first link member **210** and the second link member **220** extend, are not parallel to each other but form a certain angle. Therefore, the first link member **210** and the second link member **220** may be allowed to rotate and rapidly cope with the deformation of the body unit **100**.

(39) Hereinafter, a mobile apparatus according to the present disclosure will be described. The above descriptions of the wheel according to the present disclosure may be applied in the same manner to a wheel provided in a mobility which will be described later.

(40) A mobile apparatus according to the present disclosure may include a wheel **10** and a frame to which the wheel may be coupled.

(41) According to aspects, the wheel **10** may include a body unit **100** including a material of which a shape is reversibly deformable. the wheel may further include a first link member **210** rotatably coupled to the body unit **100**, and a second link member **220** having one side rotatably coupled to the first link member **210** and the other side rotatably coupled to the body unit **100**. A first rotary shaft **310** may be provided in a region of the body unit, in which the first link member **210** is rotatably coupled to the body unit **100**, and inserted into the first link member **210** and the body unit **100**. A second rotary shaft **320** may be provided in a region of the body unit in which the second link member **220** is rotatably coupled to the first link member **210**, and inserted into the first link member **210** and the second link member **220**. Additionally, a third rotary shaft **330** may be provided in a region of the body unit in which the second link member **220** is rotatably coupled to the body unit **100**, and inserted into the second link member **220** and the body unit **100**.

(42) According to the present disclosure, the first link member **210** and the second link member **220** may be provided adjacent to one side surface, facing the frame, among both the side surfaces of the body unit **100** in a width direction W. Such an arrangement may prevent the first link member **210** and the second link member **220** from being exposed to the outside of the mobile apparatus.

(43) A first aspect of the present disclosure provides a wheel including: a body unit including a material of which a shape is reversibly deformable; a first link member rotatably coupled to the body unit; a second link member having one side rotatably coupled to the first link member and the other side rotatably coupled to the body unit; a first rotary shaft provided in a region, in which the first link member is rotatably coupled to the body unit, and inserted into the first link member and the body unit; a second rotary shaft provided in a region, in which the second link member is rotatably coupled to the first link member, and inserted into the first link member and the second link member; and a third rotary shaft provided in a region, in which the second link member is rotatably coupled to the body unit, and inserted into the second link member and the body unit.

(44) The first rotary shaft may be provided inward from the second rotary shaft in a radial direction

(R) of the body unit.
(45) The second rotary shaft may be provided inward from the third rotary shaft in the radial direction (R).
(46) The first rotary shaft, the second rotary shaft, and the third rotary shaft may be provided in parallel to each other.
(47) The first link member and the second link member may be spaced apart from the body unit in a width direction (W) of the body unit.
(48) A distance between the first link member and the body unit may be greater than a distance between the second link member and the body unit.
(49) In a region, in which the second rotary shaft passes through the first link member and the second link member, the first link member may be inserted into the second link member.
(50) In a region, in which the second rotary shaft passes through the first link member and the second link member, the second link member may be inserted into the first link member.
(51) A first body through-hole, which the first rotary shaft is inserted into and coupled to, may be formed in the body unit, and the wheel may further include a first spacer which is inserted into and coupled to the first body through-hole to accommodate the first rotary shaft.
(52) A second body through-hole, which the third rotary shaft is inserted into and coupled to, may be formed in the body unit, and the wheel may further include a second spacer which is inserted into and coupled to the second body through-hole to accommodate the third rotary shaft.
(53) The wheel may further include a first bearing which is provided in a region, in which the first rotary shaft passes through the first link member, and accommodated inside the first link member, and the first rotary shaft may pass through the first bearing.
(54) The wheel may further include a second bearing which is provided in a region, in which the second rotary shaft passes through the first link member, and accommodated inside the first link member, and the second rotary shaft may pass through the second bearing.
(55) The first bearing may include: a first bearing inner race coupled and fixed to the first rotary shaft; and a first bearing outer race which is provided on the outside of the first bearing inner race and coupled and fixed to the first link member.
(56) The wheel may further include a first nut member coupled to the first rotary shaft, wherein the first nut member is provided on each of both sides of the body unit in a width direction (W) of the body unit.
(57) The wheel may further include a second nut member coupled to the second rotary shaft, wherein the second nut member is provided outward from the first link member in the width direction (W).
(58) The wheel may further include a third nut member coupled to the third rotary shaft, wherein the third nut member faces the second link member with the body unit therebetween in the width direction (W).
(59) The body unit may include: an inner region that forms the inside of the body unit in a radial direction (R); an outer region which is spaced outward from the inner region in the radial direction (R) and forms the outside of the body unit; and a spoke region configured to connect the inner region to the outer region, wherein the first rotary shaft is inserted into the inner region, and the third rotary shaft is inserted into the outer region.
(60) The spoke region may be provided in plurality along a circumferential direction (C) of the body unit, and each of the spoke regions may have a shape which is biased in the circumferential direction (C) toward the outside of the radial direction (R) of the body unit.
(61) In the region, in which the first link member is coupled to the second link member, a certain angle may be formed between a longitudinal direction of the first link member and a longitudinal direction of the second link member.
(62) A second aspect of the present disclosure provides a mobility (e.g., mobile apparatus) including a wheel and a frame to which the wheel is coupled, wherein the wheel includes: a body

unit including a material of which a shape is reversibly deformable; a first link member rotatably coupled to the body unit; a second link member having one side rotatably coupled to the first link member and the other side rotatably coupled to the body unit; a first rotary shaft provided in a region, in which the first link member is rotatably coupled to the body unit, and inserted into the first link member and the body unit; a second rotary shaft provided in a region, in which the second link member is rotatably coupled to the first link member, and inserted into the first link member and the second link member; and a third rotary shaft provided in a region, in which the second link member is rotatably coupled to the body unit, and inserted into the second link member and the body unit.
(63) According to the present disclosure, the twisting phenomena of the wheel may be minimized even during the turning motion of the wheel. Thus, the durability of the wheel may be enhanced, and the traveling performance of the mobile apparatus to which the wheels are mounted may be enhanced.
(64) Although the present disclosure has been described with specific exemplary embodiments and drawings, the present disclosure is not limited thereto, and it is obvious that various changes and modifications may be made by a person skilled in the art to which the present disclosure pertains within the technical idea of the present disclosure and equivalent scope of the appended claims.

Claims

1. A wheel comprising: a body comprising a material of which a shape is reversibly deformable; a first link rotatably coupled to the body; a second link having one side rotatably coupled to the first link and the other side rotatably coupled to the body; a first shaft disposed in a region, in which the first link is rotatably coupled to the body, and inserted into the first link and the body; a second shaft disposed in a region, in which the second link is rotatably coupled to the first link, and inserted into the first link and the second link; a third shaft disposed in a region, in which the second link is rotatably coupled to the body, and inserted into the second link and the body; and a first bearing disposed in the first link in a region, in which the first shaft passes through the first link, wherein the first shaft passes through the first bearing, and wherein the first bearing comprises: a first bearing inner race coupled to the first shaft; and a first bearing outer race disposed on an outside of the first bearing inner race and coupled to the first link.
2. The wheel of claim 1, wherein the first shaft is provided inward from the second shaft in a radial direction of the body.
3. The wheel of claim 2, wherein the second shaft is provided inward from the third shaft in the radial direction.
4. The wheel of claim 1, wherein the first shaft, the second shaft, and the third shaft are provided in parallel to each other.
5. The wheel of claim 1, wherein the first link and the second link are spaced apart from the body in a width direction of the body.
6. The wheel of claim 1, wherein a distance between the first link and the body is greater than a distance between the second link and the body.
7. The wheel of claim 1, wherein in a region, in which the second shaft passes through the first link and the second link, the first link is inserted into the second link.
8. The wheel of claim 1, wherein in a region, in which the second shaft passes through the first link and the second link, the second link is inserted into the first link.
9. The wheel of claim 1, wherein the first shaft is inserted into a first body through-hole formed in the body, and wherein the wheel further comprises a first spacer which is inserted into the first body through-hole to accommodate the first shaft.
10. The wheel of claim 9, wherein the third shaft is inserted into a second body through-hole formed in the body, and wherein the wheel further comprises a second spacer which is inserted into

the second body through-hole to accommodate the third shaft.
11. The wheel of claim 1, further comprising a second bearing disposed in the first link in a region, in which the second shaft passes through the first link, and wherein the second shaft passes through the second bearing.
12. The wheel of claim 1, further comprising a nut coupled to the first shaft, wherein the nut is disposed on a side of the body in a width direction of the body.
13. The wheel of claim 1, further comprising a nut coupled to the second shaft, wherein the nut is disposed outward from the first link.
14. The wheel of claim 1, further comprising a nut coupled to the third shaft, wherein the nut faces the second link with the body therebetween in a width direction.
15. The wheel of claim 1, wherein the body comprises: an inner region that forms an inside of the body in a radial direction; an outer region which is spaced outward from the inner region in the radial direction and forms an outside of the body; and a spoke region configured to connect the inner region to the outer region, wherein the first shaft is inserted into the inner region, and the third shaft is inserted into the outer region.
16. The wheel of claim 15, wherein the spoke region comprises a plurality of spokes along a circumferential direction of the body, and wherein each of the spokes has a shape which is biased in the circumferential direction toward the outside of the body in the radial direction.
17. The wheel of claim 1, wherein an angle is formed between a longitudinal direction of the first link and a longitudinal direction of the second link, and wherein the longitudinal direction of the first link and the longitudinal direction of the second link are non-parallel.
18. The wheel of claim 1, further comprising: a second bearing disposed in a region, in which the second shaft passes through the first link, wherein the second shaft passes through the second bearing.
19. A mobile apparatus comprising a wheel, wherein the wheel comprises: a body of which shape is reversibly deformable; a first link rotatably coupled to the body; a second link, wherein a first side of the second link is rotatably coupled to the first link and a second side of the second link is rotatably coupled to the body; a first shaft disposed in a region, in which the first link is rotatably coupled to the body, and inserted into the first link and the body; a second shaft disposed in a region, in which the second link is rotatably coupled to the first link, and inserted into the first link and the second link; a third shaft disposed in a region, in which the second link is rotatably coupled to the body, and inserted into the second link and the body; and a first bearing disposed in the first link in a region, in which the first shaft passes through the first link, wherein the first shaft passes through the first bearing, and wherein the first bearing comprises: a first bearing inner race coupled to the first shaft; and a first bearing outer race disposed on an outside of the first bearing inner race and coupled to the first link.
20. The mobile apparatus of claim 19, wherein the wheel further comprises a second bearing disposed in a region, in which the second shaft passes through the first link, wherein the second shaft passes through the second bearing, wherein the second bearing comprises: a second bearing inner race coupled to the second shaft; and a second bearing outer race disposed on an outside of the second bearing inner race and coupled to the first link.